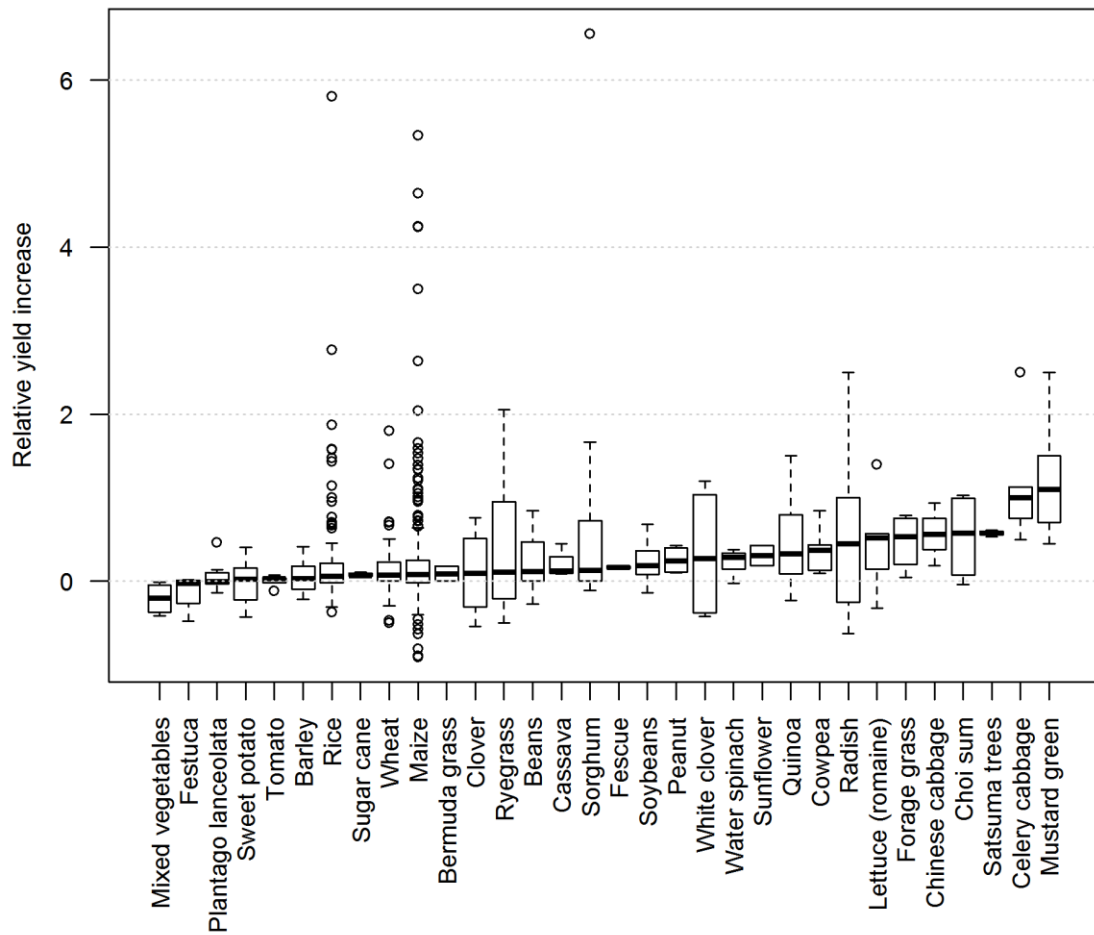
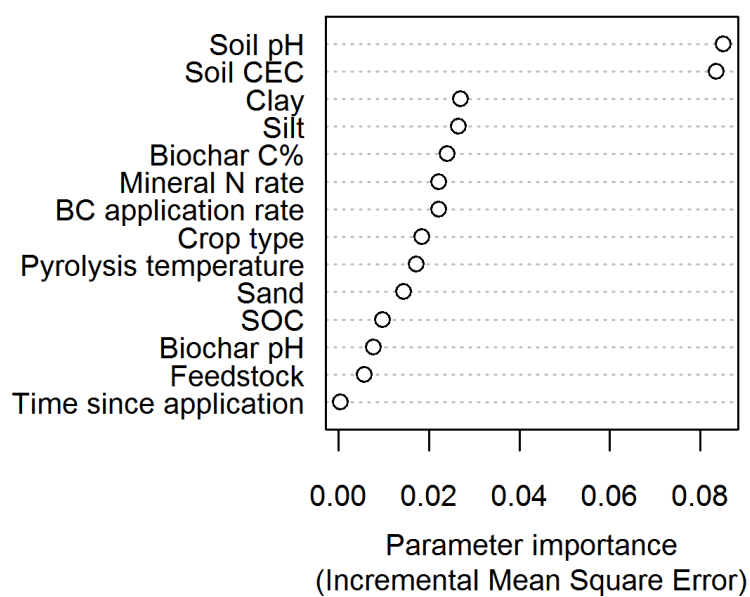
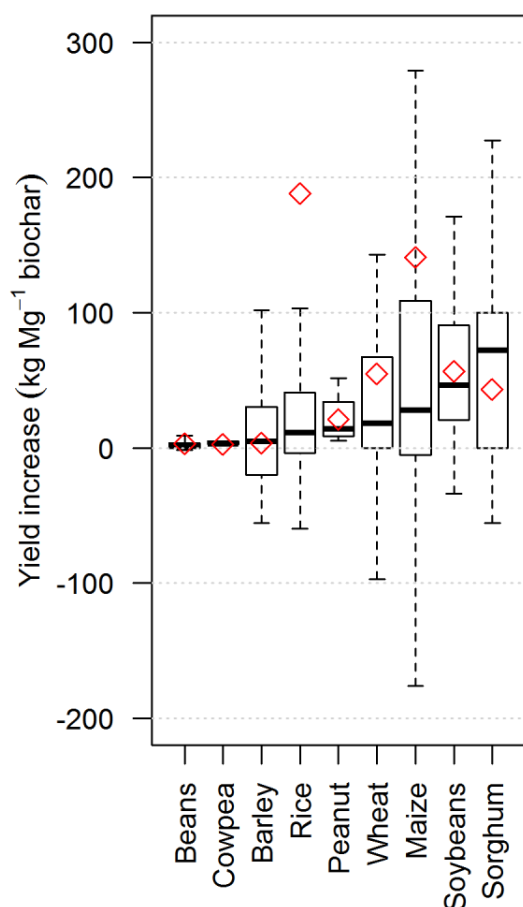




Supplementary Figure 3: Relative increase in crop yield from biochar addition. Relative increase is defined as the difference in yield between treatment and control, expressed as a fraction of control yield. Data from Refs. 4–75.



Supplementary Figure 4: Relative importance of model parameters in determining the relative crop yield increase from biochar additions. Relative importance was estimated by the incremental change in mean square error for each parameter in a random forest analysis.



Supplementary Figure 5: Incremental change in annual crop yield per Mg of applied biochar, for major field crops for which biochar studies are published. Diamonds indicate means. Outliers not shown. Note that, the ranges of biochar-yield impacts are comparable to studies indicating that, for every 1 Mg ha⁻¹ increase in SOC in the root zone of soils with low (< 2 %) initial SOC, crop yields can be increased by 20–70 kg ha⁻¹ (wheat), 10–50 kg ha⁻¹ (rice), and 30–300 kg ha⁻¹ (maize).^{76–86}

Supplementary Table 5: Production, yield and price of major food crops globally, ranked by area harvested (2008-2013 data). Yields are the global mean values. Prices are weighted means of producer prices in each country, weighted by production quantity. Data derived from FAOStat.⁸⁸

Crop	Area harvested ^a (Mha)	Production ^a (Tg)	Yield ^b (Mg ha⁻¹)	Price ^c (\$ Mg⁻¹)	Value per ha ^c (\$ ha⁻¹)
Wheat	218	713	3.3	254	828
Maize	184	1,017	5.5	239	1,318
Rice, paddy	165	746	4.5	403	1,824
Soybeans	111	276	2.5	623	1,548
Barley	50	145	2.9	223	647
Sorghum	42	61	1.5	310	452
Seed cotton	37	73	2.0	781	1,545
Rapeseed	36	73	2.0	589	1,175
Millet	33	30	0.9	383	347
Beans, dry	29	23	0.8	1,685	1,333
Sugar cane	27	1877	70.8	296	20,949
Sunflower	26	45	1.7	633	1,108
Groundnuts	25	45	1.8	1,384	2,460
Cassava	21	277	13.3	451	6,025
Potatoes	19	368	18.9	418	7,909
Oil palm	17	268	15.7	152	2,391
Chick peas	14	13	1.0	1,084	1,049
Coconuts	12	62	5.1	339	1,730
Cow peas, dry	11	6	0.5	1,762	890
Olives	10	20	2.0	1,101	2,186
Oats	10	24	2.4	194	473
Sesame seed	9	5	0.5	1,550	784
Sweet potatoes	8	111	13.4	554	7,449
Peas, dry	6	11	1.7	684	1,177
Pigeon peas	6	5	0.8	1,277	974
Pulses, nes	6	5	0.8	1,463	1,229
Rye	6	17	2.9	204	590
Yams	5	60	11.9	1,057	12,586
Sugar beet	4	250	56.3	99	5,595
Lentils	4	5	1.1	1,087	1,238
Triticale	4	15	3.8	192	728
Other Cereal	8	28	2.4	202	479

^a World total

^b World mean

^c Weighted mean (weighted by production quantity per country)

Supplementary Table 6: Interquartile ranges of the economic value of biochar yield impacts for major crops with demonstrated positive responses to biochar. Based on current (2008-2013) inter-annual and international crop price fluctuations.⁸⁸

	Yield increment (kg crop yr ⁻¹ Mg ⁻¹ C)		Crop Price (\$ Mg ⁻¹)		Value of yield increment (\$ yr ⁻¹ Mg ⁻¹ C)	
	Lower quartile	Upper quartile	Lower quartile	Upper quartile	min	max
Wheat	0	67	231	368	0.00	24.66
Maize	-5	109	224	335	-1.68	36.52
Rice	-4	41	309	623	-2.49	25.54
Sorghum	0	100	215	392	0.00	39.20
Barley	-20	31	223	647	-12.94	20.06
Peanut	9	34	714	1382	6.43	46.99
Beans	0	4	802	1,860	0.00	7.44
Cowpea	3	5	677	1,567	2.03	7.84
Soybean	21	91	510	690	10.71	62.79
Range					-12.94	62.79

Supplementary Table 7: Fraction of primary energy from low-carbon sources in RCP 2.6 ^a

	2010	2030	2050	2100
lower quartile	0.138	0.23	0.54	0.9
median	0.15	0.25	0.6	0.92
upper quartile	0.186	0.31	0.66	0.95

^a IPCC 5th Assessment Report (2014), Working Group 3, Ch. 7, Fig 7.14, p560 ⁸⁹

Supplementary Table 8: Estimated carbon intensity (Mg C GJe⁻¹) of power generation in RCP 2.6.

	2010	2030	2050	2100
mean	0.040	0.035	0.019	0.004
IQR	0.004	0.004	0.006	0.002
s.d.	0.003	0.003	0.004	0.002